TRINITY GRAMMAR SCHOOL

Mathematics Department

(NESA Student Number | Year 12 only)

2019
Year 12
Mathematics

TRIAL EXAMINATION
HSC ASSESSMENT TASK 4

Date of Assessment Task: Thursday, 22 August 2019

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Write your NESA Student Number and your Class teacher on the question paper and on any answer sheets or writing booklets used to write your responses to the questions submitted
- If you do not attempt a question you must submit an answer sheet or writing booklet for that question clearly indicating N/A and your NESA Student Number or Name

Total marks: 100

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6 – 14)

- Attempt Questions 11 – 16
- Allow about 2 hour and 45 minutes for this section

- **HSC Assessment Weighting:** 40%

Section I

10 marks
Attempt Questions 1 – 10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

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1 What is the solution to the equation $4^x = 32$?

(A) 0.4

(B) 2.5

(C) 3

(D) 8

2 What values of x is the curve $f(x) = 2x^3 + x^2$ concave down?

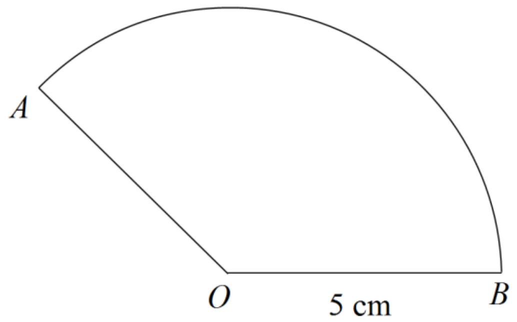
(A) $x < -\frac{1}{6}$

(B) $x > -\frac{1}{6}$

(C) $x < -6$

(D) $x > 6$

3 *AOB* is a sector of a circle, centre *O* and radius 5 cm. The sector has an area of 10π cm^2.



Not to scale

What is the arc length of the sector?

(A) 2π

(B) 4π

(C) 6π

(D) 10π

4 A can of soup is in the shape of a closed cylinder with a vertical height *h* cm and a radius *r* cm. The volume of the can of soup is 400 cm^3. What is the radius of the can if the surface area of the metal used to make the can of soup is to be minimized?

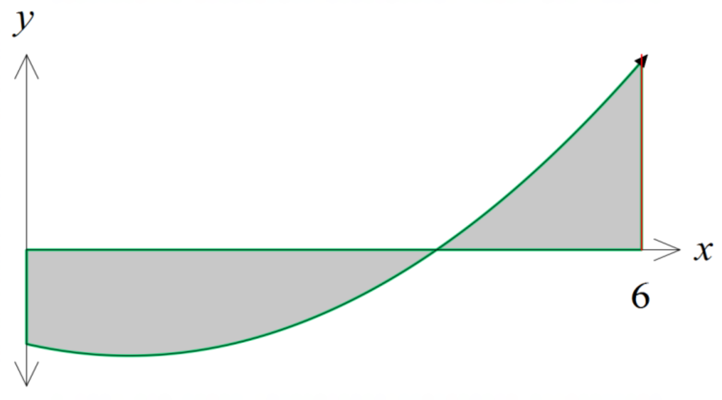
(A) $\sqrt[3]{\frac{100}{\pi}}$

(B) $\sqrt[3]{\frac{200}{\pi}}$

(C) $\sqrt[3]{\frac{400}{\pi}}$

(D) $\sqrt[3]{\frac{800}{\pi}}$

5 The diagram below shows the graph of $y = x^2 - 2x - 8$.



What is the correct expression for the area bounded by the *x*-axis and the curve $y = x^2 - 2x - 8$ between $0 \le x \le 6$?

(A) $A = \int_0^5 x^2 - 2x - 8\,dx + \left| \int_5^6 x^2 - 2x - 8\,dx \right|$

(B) $A = \int_0^4 x^2 - 2x - 8\,dx + \left| \int_4^6 x^2 - 2x - 8\,dx \right|$

(C) $A = \left| \int_0^5 x^2 - 2x - 8\,dx \right| + \int_5^6 x^2 - 2x - 8\,dx$

(D) $A = \left| \int_0^4 x^2 - 2x - 8\,dx \right| + \int_4^6 x^2 - 2x - 8\,dx$

6 What is the value of $\sum_{r=1}^{3} 2^{1-r}$?

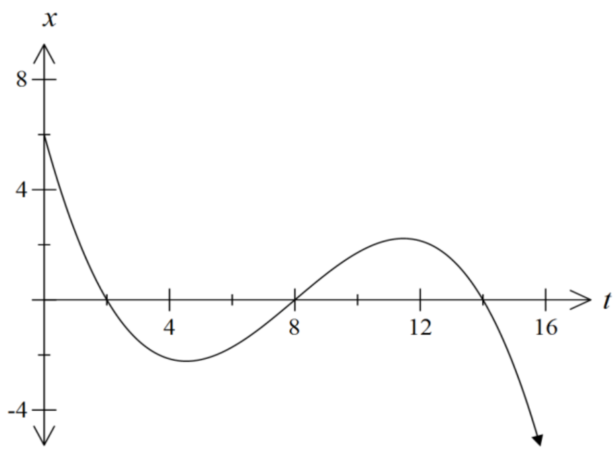
(A) $\frac{3}{4}$

(B) $1\frac{3}{4}$

(C) 6

(D) 7

7 The displacement, x metres, from the origin of a particle moving in a straight line at any time (t seconds) is shown in the graph.



When was the particle at rest?

(A) $t = 4.5$ and $t = 11.5$

(B) $t = 0$

(C) $t = 2$, $t = 8$ and $t = 14$

(D) $t = 1.5$ and $t = 8$

8 What is the solution to the equation $\cos 2x = \frac{1}{2}$ in the domain $-\pi \le x \le \pi$?

(A) $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{-5\pi}{6}, \frac{-\pi}{6}$

(B) $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{-11\pi}{12}, \frac{-\pi}{12}$

(C) $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

(D) $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}$

9 The fourth term of an arithmetic series is 27 and the seventh term is 12. What is the common difference?

(A) -5

(B) 5

(C) 13

(D) 42

10 What is the solution to the equation $\log_e(x+2)-\log_e x=\log_e 4$?

(A) $\frac{2}{5}$

(B) $\frac{2}{3}$

(C) $\frac{3}{2}$

(D) $\frac{5}{2}$

Section II

90 marks
Attempt Questions 11 – 16
Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.
In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Marks

Question 11 (15 marks) **[Use a SEPARATE writing booklet]**

(a) Evaluate e^3 correct to one significant figure. **2**

(b) Factorise $4x^2 + 7x - 2$. **2**

(c) Simplify $1 - \dfrac{1}{x+1}$. **2**

(d) Solve $|3x - 1| = 8$. **2**

(e) Rationalise the denominator of $\dfrac{\sqrt{2}}{\sqrt{2} - 1}$. **2**

(f) Find the sum to 18 terms of the arithmetic series $5 + 3 + 1 + \ldots\ldots\ldots$ **2**

(g) Solve $f'(x) = 0$, where $f(x) = \dfrac{e^{3x}}{x^2}$. **3**

End of Question 11

Marks

Question 12 (15 marks) **[Use a SEPARATE writing booklet]**

(a) Differentiate with respect to x.

(i) $(x^2 - 4)^3$ **2**

(ii) $x^2 \sin x$ **2**

(iii) $\dfrac{\ln x}{x}$ **2**

(b) (i) Find $\int e^{3x-4}\,dx$. **1**

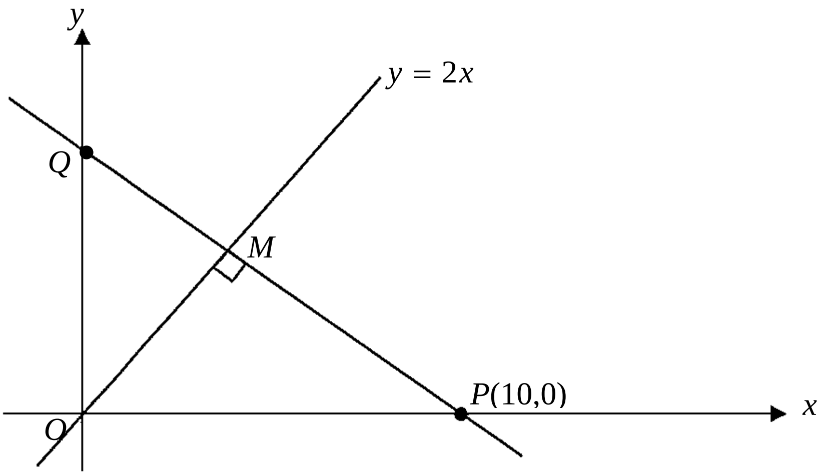
(ii) Find $\int \tan^2 x\,dx$. **2**

(iii) Evaluate $\displaystyle\int_1^2 \frac{x^2 - x}{x^2}\,dx$. **3**

(c) Find the volume of the solid formed when the curve $y = \dfrac{2}{\sqrt{2x-1}}$ is rotated $360°$ about the x-axis from $x = 1$ to $x = 5$, giving an exact answer in terms of π. **3**

End of Question 12

Question 13 (15 marks) **[Use a SEPARATE writing booklet]** **Marks**



NOT TO SCALE

In the above diagram the equation of the line *OM* is $y = 2x$. The line *MP* is perpendicular to *OM*.

(a) (i) Find the gradient of the line *MP*. **1**

(ii) Show that the equation of the line *MP* is $x + 2y - 10 = 0$. **2**

(iii) Find the coordinates of the point *M*. **1**

(b) The vertex of a parabola is $V(1, 2)$ and its directrix is the line $y = -2$

(i) Sketch the parabola. **1**

(ii) State the focal length of the parabola **and** write down the coordinates of its focus. **2**

(iii) Find the Cartesian equation of the parabola. **1**

(c) (i) Sketch the graph of $y = \ln 2x$. **1**

(ii) Find the **exact** area bounded by the curve $y = \ln 2x$, the lines $y = 1$ and $y = 2$ and the y – axis. **2**

Question 13 continues on page 9

Question 13 continued

(d) For the curve $y = -3\cos 2x$:

(i) state the amplitude. **1**

(ii) state the period. **1**

(iii) Hence, sketch **and** label the coordinates intercepts of the graph over the domain $0 \le x \le 2\pi$. **2**

End of Question 13

Marks

Question 14 (15 marks) **[Use a SEPARATE writing booklet]**

(a) A population of insects is growing so that the number of insects is given by the formula $N = 8000 \times e^{0.2t}$ where t is the time in days from now.

(i) Find the population after 5 days has elapsed. **2**

(ii) An eradication program will begin when the **rate of growth** exceeds 25 000 insects / day. In which day will the eradication program begin? **3**

(b) If α and β are the roots of the quadratic equation $x^2 - 2x - 1 = 0$, **3**

find the value of

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\alpha^2 + \beta^2$

(c) Consider the curve $f\left(x\right) = 12x^2 - x^4$.

(i) Find all stationary points and determine their nature. **4**

(ii) Show that $f(x)$ is an even function. **1**

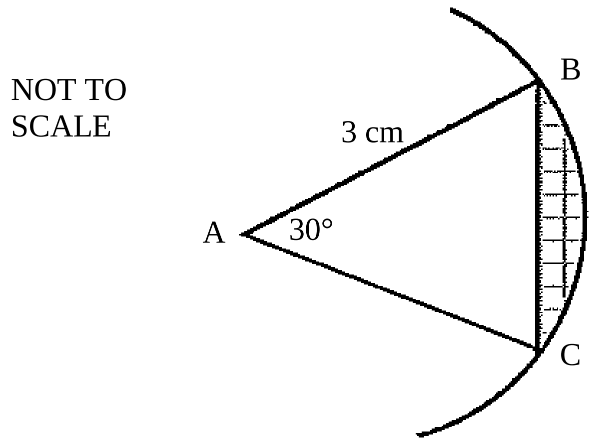
(iii) Sketch the curve for $-3 \le x \le 3$, showing the stationary points, and the x and y intercepts. **2**

End of Question 14

Marks

Question 15 (15 marks) **[Use a SEPARATE writing booklet]**

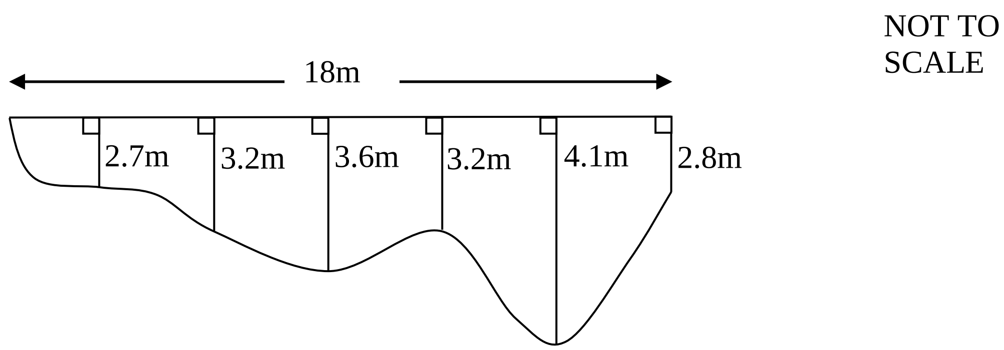
(a) In the diagram $\angle BAC = 30^o$ and a circular arc of radius 3 cm with centre A is constructed from point B to point C.



(i) Find the area of ΔABC. **1**

(ii) Calculate the **exact** area of the shaded segment. **2**

(b) A surveyor measures the depth of a river at equal intervals across its width. The river is 18m wide and the measurements are as follows:



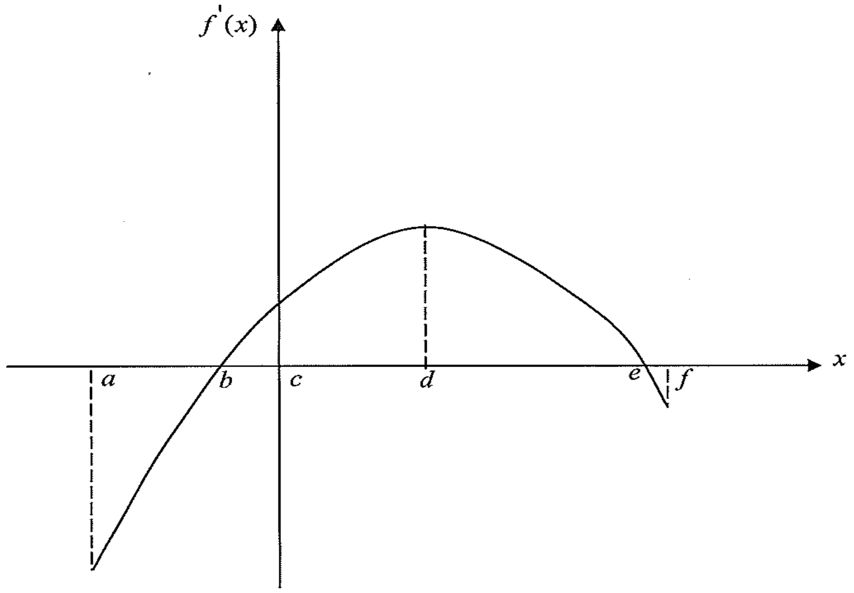
(i) Using the trapezoidal rule and seven function values, find the approximate cross-sectional area of the river. **3**

(ii) Water flows through this section of the river at a speed of 2m/s. Calculate the approximate volume of water that flows past this section in one hour. **1**

Question 15 continues on page 12

Question 15 continued

(c) Given the function $y = f(x)$, the graph below represents its **gradient function** i.e. $y = f'(x)$



From the information given, there are two possible x values in the domain $a \le x \le f$ at which $f(x)$ could have its largest values. Write down these two values. **2**

(d) The displacement of a particle moving along the x-axis is given by

$$x = t - \frac{1}{1+t},$$

where x is the displacement from the origin in metres, t is the time in seconds, and $t \ge 0$.

(i) Show that the acceleration of the particle is always negative. **2**

(ii) What value does the velocity approach as t increases indefinitely? **1**

(e) Prove the identity: **3**

$$\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta} = 2\sec^2\theta$$

End of Question 15

Marks

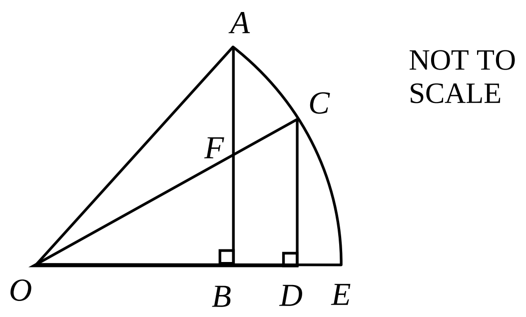
Question 16 (15 marks) **[Use a SEPARATE writing booklet]**

(a) A pendulum is set swinging. It travels 27 cm in the first swing, 18 cm in the second swing, and 12 cm in the third swing and so on in the same pattern.

(i) Find the total distance covered after 8 swings, correct to one decimal place. **2**

(ii) Find the total distance covered by the pendulum before it comes to rest. **2**

(b) In the diagram, ACE is an arc of a circle with centre O and radius $2\sqrt{13}$ cm.
$\angle OBA = \angle ODC = 90°$, $OB = CD = 4$ cm.



Copy this diagram into your writing booklet.

(i) Prove that ΔOAB is congruent to ΔOCD. **3**

(ii) Find the length of BD. **1**

(iii) ΔFOB is similar to ΔCOD. (Do NOT show this.)
Find the area of $BDCF$. **2**

Question 16 continues on page 14

Question 16 continued

(c) A fund is set up with a single investment of \$2000 to provide for an **annual** mathematics prize of \$150. The fund accrues interest at 5% p.a. paid yearly and the first prize is awarded one year after the investment is made.

Let V_n be the value of the fund at end of n years.

(i) Show that $V_3 = 2000(1.05)^3 - 150\left[1.05^2 + 1.05 + 1\right]$ **2**

(ii) Write down an expression for V_n, the value after n years. **1**

(iii) Find the number of years for which the full prize can be awarded. **2**

End of Question 16
End of Task

2019 12MA TASK 4 TRIAL HSC - Solutions

Section 1

1	B	6	B
2	A	7	A
3	B	8	A
4	B	9	A
5	D	10	B

Section 2
Question 11

a) $e^3 = 20$

(Bald incorrect answer zero.)

b) $(4x-1)(x+2)$

c) $1 - \dfrac{1}{x+1} = \dfrac{x+1-1}{x+1}$

$= \dfrac{x}{x+1}$

d) $|3x-1| = 8$

$3x-1 = 8$ or $3x-1 = -8$

$3x = 9$ $\quad$ $3x = -7$

$x = 3$ $\quad$ $x = -\dfrac{7}{3}$

e) $\dfrac{\sqrt{2}}{\sqrt{2}-1} = \dfrac{\sqrt{2}}{\sqrt{2}-1} \times \dfrac{\sqrt{2}+1}{\sqrt{2}+1}$

$= \dfrac{\sqrt{2}(\sqrt{2}+1)}{2-1}$

$= 2 + \sqrt{2}$

f) $S_{18} = \frac{18}{2}\left[2 \times 5 + 17 \times -2\right]$

$= 9(10 - 34)$

$= -216$

g) $f(x) = \frac{e^{3x}}{x^2}$

$$f'(x) = \frac{x^2 \times 3e^{3x} - e^{3x} \times 2x}{(x^2)^2}$$

$$= \frac{xe^{3x}(3x - 2)}{x^4}$$

If $f'(x) = 0$ then $3x - 2 = 0$

$3x = 2$

$x = \frac{2}{3}$

<u>Question 12</u>

a) (i) $\frac{d}{dx}(x^2 - 4)^3 = 3(x^2 - 4)^2 \times 2x$

$= 6x(x^2 - 4)^2$

(ii) $\frac{d}{dx}(x^2 \sin x) = \sin x \times 2x + x^2 \cos x$

$= 2x \sin x + x^2 \cos x$

(iii) $\frac{d}{dx}\left(\frac{\ln x}{x}\right) = \frac{x \times \frac{1}{x} - \ln x \times 1}{x^2}$

$= \frac{1 - \ln x}{x^2}$

b) (i) $\int e^{3x-4}\,dx = \frac{1}{3}e^{3x-4} + c$

(ii) $\int \tan^2 x \,dx = \int \sec^2 x - 1 \,dx$

$= \tan x - x + c$

(iii) $\int_1^2 \frac{x^2 - x}{x^2}\,dx = \int_1^2 1 - \frac{1}{x}\,dx$

$= \left. x - \ln x \right]_1^2$

$= 2 - \ln 2 - (1 - \ln 1)$

$= 1 - \ln 2$

c) $V = \pi \int_1^5 \left(\frac{2}{\sqrt{2x-1}}\right)^2 dx$

$= \pi \int_1^5 \frac{4}{2x-1}\,dx$

$= \left. 2\pi \ln(2x-1) \right]_1^5$

$= 2\pi \ln 9 - 2\pi \ln 1$

$= 2\pi \ln 9$ units^3

<u>Question 13</u>

a) (i) $m_{mp} = -\frac{1}{2}$

(ii) $y - 0 = -\frac{1}{2}(x - 10)$

$y = -\frac{1}{2}x + 5$

$2y = -x + 10$

$x + 2y - 10 = 0$

(iii) $x + 2y - 10 = 0$ ①

$y = 2x$ ②

Sub ② into ①

$x + 4x - 10 = 0$

$5x = 10$

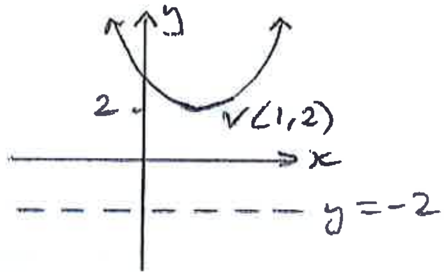
$x = 2$

Sub $x = 2$ into ②

$y = 2(2) = 4$

$\therefore$ M is $(2, 4)$

b) (i)



(ii) $a = 4$ units

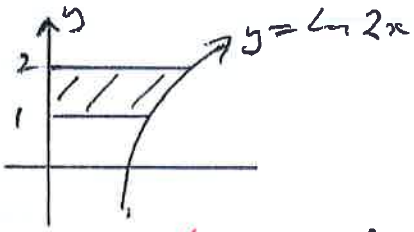
Focus $= (1, 6)$

(iii) $(x-h)^2 = 4a(y-k)$ $\quad a=4,\ V$ is $(1,2)$

$(x-1)^2 = 4(4)(y-2)$

$(x-1)^2 = 16(y-2)$

c)



$y = \ln 2x$

$e^y = 2x$

$x = \frac{1}{2}e^y$

$A = \int_1^2 \frac{1}{2}e^y\,dy = \frac{1}{2}e^y\Big]_1^2$

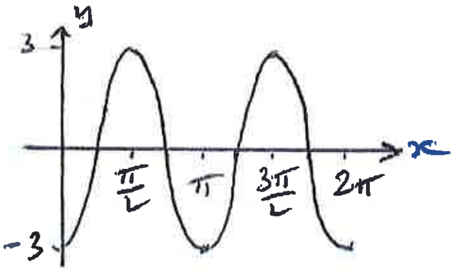
$= \frac{1}{2}(e^2 - e)$

$= \underline{\frac{e(e-1)}{2}}$ units^2

d) (i) amplitude $= 3$

(ii) Period $= \frac{2\pi}{2} = \pi$

(iii)



correct amplit.
correct cut pts.

<u>Question 14</u>

a) (i) $N = 8000e^{0.2\times5}$

$= 8000e$

$= 21746$

(ii) $N = 8000e^{0.2t}$

$\frac{dN}{dt} = 0.2 \times 8000e^{0.2t}$

$= 1600e^{0.2t} > 25000$

$e^{0.2t} > \frac{250}{16}$

$0.2t > \ln\left(\frac{250}{16}\right)$

$t > \frac{\ln\left(\frac{250}{16}\right)}{0.2}$

$t > 13.744$

$\therefore$ Eradication begins on day 14.

b) (i) $\alpha + \beta = \frac{-b}{a}$

$= \frac{2}{1}$

$= 2$ ✓

(ii) $\alpha\beta = \frac{c}{a}$

$= \frac{-1}{1}$

$= -1$ ✓

(iii) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

$= 2^2 - 2(-1)$

$= 4 + 2$

$= 6$ ✓

c) $f(x) = 12x^2 - x^4$

$f'(x) = 24x - 4x^3 = 0$ for stat. pts ✓

$4x(6 - x^2) = 0$

$x = 0, \pm\sqrt{6}$.

When $x = 0$, $y = 0$

Now test $(0,0)$

$f''(0) = 24 - 12(0)^2$

$= 24 > 0$

$\therefore (0,0)$ is a min ✓

When $x = \sqrt{6}$, $y = 12(6) - 36$

$= 36$

Now test $(\sqrt{6}, 36)$

$f''(\sqrt{6}) = 24 - 12(\sqrt{6})^2$

$= -48 < 0$

$\therefore (\sqrt{6}, 36)$ is a max. ✓

When $x = -\sqrt{6}$, $y = 12(6) - 36$

$= 36$

Now test $(-\sqrt{6}, 36)$

$f''(-\sqrt{6}) = 24 - 12(6)$

$= -48 < 0$

$\therefore (-\sqrt{6}, 36)$ is a max ✓

(ii) $f(-x) = 12(-x)^2 - (-x)^4$
$= 12x^2 - x^4$
$= f(x)$

$\therefore f(x)$ is an even function

(iii)

1 - shape
1 - points

Question 15

a) (i) $A = \frac{1}{2} \times 3^2 \times \sin 30°$
$= \frac{9}{4}$ cm^2

(ii) $A = \frac{1}{2} r^2 (\theta - \sin\theta)$

$= \frac{1}{2} \times 3^2 \left(\frac{\pi}{6} - \sin\frac{\pi}{6}\right)$

$= \frac{9}{2}\left(\frac{\pi}{6} - \frac{1}{2}\right)$

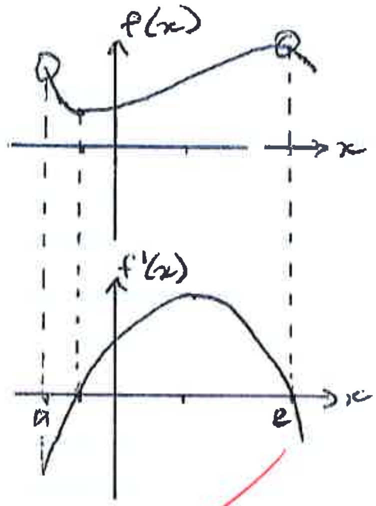
$= \frac{3(\pi - 3)}{4}$ cm^2

b) (i) $A \doteq \frac{3}{2}\left[0 + 2(2.7 + 3.2 + 3.6 + 3.2 + 4.1) + 2.8\right]$

$= 54.6 m^2$

(ii) $V = 54.6 \times 2$ m^3/s
$= 54.6 \times 2 \times 60 \times 60$ m^3/h
$= 393120$ m^3

c)



$x = a$ and $x = e$

d) (i) $v = \dfrac{dx}{dt} = 1 + \dfrac{1}{(1+t)^2}$

$a = \dfrac{d^2x}{dt^2} = \dfrac{-2}{(1+t)^3} < 0$ as $(1+t)^3 > 0$, for $t \geq 0$

(ii) $\lim\limits_{t \to \infty} \left[1 + \dfrac{1}{(1+t)^2} \right] = 1$

since $\dfrac{1}{(1+t)^2} \to 0$ as $t \to \infty$

$\therefore$ Velocity $\to 1$ as t becomes large

e) $LHS = \dfrac{1}{1+\sin\theta} + \dfrac{1}{1-\sin\theta}$

$= \dfrac{1-\sin\theta + 1+\sin\theta}{(1+\sin\theta)(1-\sin\theta)}$

$= \dfrac{2}{1-\sin^2\theta}$

$= \dfrac{2}{\cos^2\theta}$

$= 2\sec^2\theta$

$= RHS$.

__Question 16__

a) Distance $= 27 + 18 + 12 + \dots$

$a = 27, \quad r = \frac{18}{27} = \frac{2}{3}, \quad n = 8$

(i) $S_8 = \dfrac{27\left[1 - \left(\frac{2}{3}\right)^8\right]}{1 - \frac{2}{3}}$

$= 77.8\text{ cm}$

(ii) $S_\infty = \dfrac{a}{1-r}$

$= \dfrac{27}{1 - \frac{2}{3}}$

$= 81\text{ cm}$

b) (i) In ΔOAB and ΔOCD

$OA = OC$ (radii of a circle are equal)

$OB = CD$ (Given)

$\angle OBA = \angle ODC$ (Given) 2

$\therefore \Delta OAB \equiv \Delta OCD$ (RHS test)

(ii) $(2\sqrt{3})^2 = OD^2 + 4^2$

$OD = 6\text{ cm}$

$BD = 6 - 4 = 2\text{cm}$

(iii) $\dfrac{FB}{CD} = \dfrac{OB}{OD}$

$FB = \dfrac{4 \times 4}{6} = 2\frac{2}{3}\text{ cm}$

$A = \frac{1}{2} \times 2 \times \left(2\frac{2}{3} + 4\right)$

$= 6\frac{2}{3}\text{ cm}^2$

c)(i) $V_1 = 2000(1.05) - 150$

$V_2 = V_1 \times 1.05 - 150$

$= 2000(1.05)^2 - 150 \times 1.05 - 150$

$V_3 = V_2 \times 1.05 - 150$

$= 2000(1.05)^3 - 150(1.05)^2 - 150(1.05) - 150$

$= 2000(1.05)^3 - 150[1.05^2 + 1.05 + 1]$

(ii) $V_n = 2000(1.05)^n - 150[1.05^{n-1} + \dots + 1.05^2 + 1.05 + 1]$

(iii) $V_n = 2000(1.05)^n - 150 \times S_n$

$S_n = \dfrac{1.05^n - 1}{1.05 - 1}$

$= 20(1.05^n - 1)$

$\therefore V_n = 2000(1.05)^n - 150 \times 20(1.05^n - 1)$

$= 2000(1.05)^n - 3000(1.05^n - 1)$

but $V_n = 0$

$\therefore 0 = 2000(1.05)^n - 3000(1.05^n - 1)$

$0 = 2000(1.05)^n - 3000(1.05)^n + 3000$

$1000(1.05)^n = 3000$

$1.05^n = 3$

$n = \dfrac{\ln 3}{\ln 1.05} = 22.5$ years

$\therefore$ The prize can be awarded for 22 years.